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CSC 4035

EXERCISE 1

Part A – Primary & Shadow Rays

(a) Compute diffuse and specular RGB before ambient.

$$\begin{aligned}\text{diffuse} &= k_d * (I \cdot n) \\ &= (0,1,0) \cdot 0.8 \\ &= (0,0.8,0) \\ \text{specular} &= k_s * (h \cdot n)^n \\ &= 1 * (0.6)^{20} \\ &= 0.00003656\end{aligned}$$

(b) Add ambient (ka) and give final RGB.

$$\begin{aligned}\text{ambient} &= k_a * k_d \\ &= 0.05 * (0,1,0) \\ &= (0, 0.05, 0) \\ \text{final RGB} \\ &= ((0, 0.8, 0) + (0.00003656, 0.00003656, 0.00003656)) + (0, 0.05, 0) \\ &= (0.00003656 + 0, 0.05 + 0.80003656, 0 + 0.00003656) \\ &= (0.00003656, 0.85003656, 0.00003656)\end{aligned}$$

2. Shadow Test

Since the shadow ray from P to L does not pass through any objects before arriving at the light source, P is not in shadow.

Part B – Reflection Ray

(a) Write formula for reflection direction r about plane normal n = (0,1,0).

Given incident direction d and normal n:

$$r = d - 2(d \cdot n) n$$

(b) Determine if reflected ray hits Object A (given hit $R = (0.6, 0.2, -4.4)$).

Yes, the reflected ray hits Object A at $R = (0.6, 0.2, -4.4)$, which is inside the sphere (center $(0, 0, -6)$, radius 1).

(c) Explain (≤ 3 sentences) how 'TraceRay' combines mirrored colour with plane's material. TraceRay recursively traces the reflection ray and multiplies the returned colour by the surface's specular coefficient (k_s). For a perfect mirror ($k_s = 1$), it uses the reflected colour directly. The final colour combines this with any local shading (diffuse/specular) from the surface itself.

Part C – Refraction Concept Check

1. In 3–4 sentences, describe the path of the ray as it enters the sphere, refracts, and exits toward the eye. Explicitly state (i) whether the ray bends toward or away from the surface normal on entry, and (ii) whether θ_i is greater than, less than, or equal to θ_t .

When the ray enters the glass sphere from air, it bends towards the normal because it's entering a denser medium. Inside the sphere, it travels in a straight line until it strikes the opposite surface. When it exits the glass into air, it bends away from the normal because it's traveling to a less dense medium. By Snell's Law, the angle of exit is the same as the original angle of incidence.

2. State Snell's Law and decide whether total internal reflection can occur (justify briefly).
 - $n_1 \cdot \sin(\theta_i) = n_2 \cdot \sin(\theta_t)$
 - Yes, when the ray tries to exit from glass to air at an angle greater than the critical angle

Part D – Soft Thought Experiment

Describe algorithmic change in shadow step to produce soft edges.

Instead of tracing one shadow ray to a point light, we:

- Sample a number of points on the area light
- For each sample, test visibility from the surface point
- Count the number of samples that are not occluded (N_{visible})
- Shadow attenuation = $N_{\text{visible}} / \text{total_samples}$

2. For P from Part A, 4 samples: 2 unoccluded, 2 blocked.

- Compute shadow attenuation factor.

attenuation factor = $2 / 4 = 0.5$

Final RGB = $0.5 \cdot (0.64, 0.64, 0.64) = (0.32, 0.32, 0.32)$
 $0.5 \cdot (0.64, 0.64, 0.64) = (0.32, 0.32, 0.32)$

- Give resulting RGB if fully-lit colour = $(0.64, 0.64, 0.64)$.

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Part E – Compact Pseudocode

Fill in the missing lines (1–4) of the “business-card-sized” ray tracer:

function TraceRay(ray, depth):

 if depth > MAX_DEPTH: return (0,0,0)

 hit = intersectScene(ray)

 if hit == None: return (0,0,0) // background

 colour = Shade(hit) // primary shading

 material = hit.material

 // 1. if material. ks > 0: // spawn reflection?

 // 2. reflected_colour = TraceRay(reflected_ray, depth+1)

 // 3. colour += material's * reflected_colour // combine with ks / kt

 // 4. return colour // return final RGB

function Shade(hit):

 // compute Blinn-Phong + ambient + shadow attenuation